



AQ7.2: Activity Questions 2 - Not Graded



This assignment will not be graded and is only for practice.

Level 1

1 point

Let L be an affine subspace of $\mathbb{R}^3$ defined as $L = \{(x, y, z) \mid x + y + 2z = 6\}$. Then which of the following subspaces of $\mathbb{R}^3$ corresponds to the affine subspace L ?

☐

$$\{(x, y, z) \mid x + y + z = 5\}$$

☐

$$\{(x, y, z) \mid x + y + 2z = 0\}$$

☐

$$\{(x, y, z) \mid x + 2y + z = 0\}$$

☐

$$\{(x, y, z) \mid 2x + y + z = 0\}$$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$\{(x, y, z) \mid x + y + 2z = 6\}$$

1 point

Which of the following subsets of $\mathbb{R}^2$ represent an affine subspace of $\mathbb{R}^2$?

☐

$$\{(x, y) \mid x^2 + y^2 = 0\}$$

☐

$$\{(x, y) \mid x^2 + y^2 = 4\}$$

☐

$$\{(x, y) \mid x^5 + 1 = y\}$$

☐

$$\{(x, y) \mid 3x + y = 1\}$$

☐

$$\{(x, y) \mid x = 1\}$$

☐

$$\{(x, y) \mid y = 5x^2\}$$

☐

$$\{(x, y) \mid y = x^3 + 2\}$$

☐

$$\{(1, 2)\}$$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$\{(x, y) \mid x^2 + y^2 = 0\}$$

$$\{(x, y) \mid 3x + y = 1\}$$

$$\{(x, y) \mid x = 1\}$$

$$\{(1, 2)\}$$

1 point



Which of following subsets of R^3 represent an affine subspace of R^3 ?

☐

$$\{(x, y, z) \mid x^2 + y^2 + z^2 = 0\} \{(x, y, z) \mid x^2 + y^2 + z^2 = 0\}$$

☐

$$\{(x, y, z) \mid x^2 + z^2 + y = 1\} \{(x, y, z) \mid x^2 + z^2 + y = 1\}$$

☐

$$\{(x, y, z) \mid x + y + z = 2\} \{(x, y, z) \mid x + y + z = 2\}$$

☐

$$\{(x, y, z) \mid 2x + y = 3z\} \{(x, y, z) \mid 2x + y = 3z\}$$

☐

$$\{(x, y, z) \mid xy = z\} \{(x, y, z) \mid xy = z\}$$

☐

$$\{(x, y, z) \mid y = x + z\} \{(x, y, z) \mid y = x + z\}$$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$\{(x, y, z) \mid x^2 + y^2 + z^2 = 0\} \{(x, y, z) \mid x^2 + y^2 + z^2 = 0\}$$

$$\{(x, y, z) \mid x + y + z = 2\} \{(x, y, z) \mid x + y + z = 2\}$$

$$\{(x, y, z) \mid 2x + y = 3z\} \{(x, y, z) \mid 2x + y = 3z\}$$

$$\{(x, y, z) \mid y = x + z\} \{(x, y, z) \mid y = x + z\}$$

1 point

Let U be a subspace of the vector space R^3 and a basis of U is given by $\{(0, 1, -3), (1, 0, -1)\}$ $\{(0, 1, -3), (1, 0, -1)\}$. Then which of the following subsets of R^3 is an affine subspace of R^3 such that the corresponding vector subspace is U ?

☐

$$L = \{(x, y, z) \mid x + 3y + z = 3\} L = \{(x, y, z) \mid x + 3y + z = 3\}$$

☐

$$L = \{(x, y, z) \mid x + 3y + 2z = 0\} L = \{(x, y, z) \mid x + 3y + 2z = 0\}$$

☐

$$L = \{(x, y, z) \mid x + 3y + z = 0\} L = \{(x, y, z) \mid x + 3y + z = 0\}$$

☐

$$L = \{(x, y, z) \mid x + 3y + z = 8\} L = \{(x, y, z) \mid x + 3y + z = 8\}$$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$L = \{(x, y, z) \mid x + 3y + z = 3\} L = \{(x, y, z) \mid x + 3y + z = 3\}$$

$$L = \{(x, y, z) \mid x + 3y + z = 0\} L = \{(x, y, z) \mid x + 3y + z = 0\}$$

$$L = \{(x, y, z) \mid x + 3y + z = 8\} L = \{(x, y, z) \mid x + 3y + z = 8\}$$

1 point

Which of the following mappings is an affine mapping from R^2 to an affine subspace of R^3 ?

☐

$$T(x, y) = (x + 1, y, xy) T(x, y) = (x + 1, y, xy)$$

☐

$$T(x, y) = (x, y, x + y + 5) T(x, y) = (x, y, x + y + 5)$$

☐

$$T(x, y) = (2x - y + 2, 3y + 3, x + y - 1) T(x, y) = (2x - y + 2, 3y + 3, x + y - 1)$$

☐

$$T(x, y) = (3x, y^2, x) T(x, y) = (3x, y^2, x)$$

☐

$$T(x, y) = (x, x + y, 0) \quad T(x, y) = (x, x + y, 0)$$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$T(x, y) = (x, y, x + y + 5) \quad T(x, y) = (x, y, x + y + 5)$$

$$T(x, y) = (2x - y + 2, 3y + 3, x + y - 1) \quad T(x, y) = (2x - y + 2, 3y + 3, x + y - 1)$$

$$T(x, y) = (x, x + y, 0) \quad T(x, y) = (x, x + y, 0)$$

Level 2

1 point

Consider a system of linear equations

$$\begin{aligned} -x + y - z &= 1 \\ x - y + z &= -1 \\ x + z &= 0 \end{aligned} \quad \begin{aligned} -x + y - z &= 1 \\ x - y + z &= -1 \\ x + z &= 0 \end{aligned}$$

Which of the following is an affine subspace L of R^3 such that if $v \in L$, then v is a solution of the system of linear equations?

☐

$$\{(t, -1, t) \mid t \in R\}$$

☐

$$\{(-t, 1, t) \mid t \in R\}$$

☐

$$\{(-t, 1, -t) \mid t \in R\}$$

☐

$$\{(-t, -1, t) \mid t \in R\}$$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$\{(-t, 1, t) \mid t \in R\}$$

1 point

Consider the following affine mappings from R^3 to an affine subspace of R^3

$$f_1(x, y, z) = (x + 1, x + 2y - 2, x + y + 2z) \quad f_1(x, y, z) = (x + 1, x + 2y - 2, x + y + 2z)$$

$$f_2(x, y, z) = (x + 5, x + 2y - 3, x + y + 2z + 8) \quad f_2(x, y, z) = (x + 5, x + 2y - 3, x + y + 2z + 8)$$

$$f_3(x, y, z) = (2x + 4, y + z - 1, x + y + 5) \quad f_3(x, y, z) = (2x + 4, y + z - 1, x + y + 5)$$

Which of the following option(s) is(are) true?

☐

A linear transformation corresponding to f_1 is $T(x, y, z) = (x, x + 2y, x + y + 2z)$

☐

The linear transformations corresponding to f_1 and f_2 are not the same.

☐

A linear transformation corresponding to f_3 is $T(x, y, z) = (x, y + z, x + y)$

☐

A linear transformation corresponding to f_3 is $T(x, y, z) = (2x, y + z, x + y)$

No, the answer is incorrect.

Score: 0

Accepted Answers:

A linear transformation corresponding to f_1 is $T(x, y, z) = (x, x + 2y, x + y + 2z)$

A linear transformation corresponding to f_3 is $T(x, y, z) = (2x, y + z, x + y)$

Suppose $f: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is an affine mapping such that $f(2, 3) = (3, -1)$, $f(2, 1) = (1, 2)$ and $f(1, 0) = (0, 1)$. If $f(-2, 5) = (a, b)$, then value of $a + b$ is

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) -9

1 point

[Check Answers](#)

Your score is: 0/8